

New ThaaT System for Hindustaanee Raags

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1. Introduction

The classification of raags into thaaTs is a task highly relevant to Hindustaanee raag music pedagogy and musicology, even if not to performance practice. However, the widely prevalent Bhatkhande 10 thaaT system is oft criticized for its limited applicability to certain raags, its lack of internal coherence and consistency, and its potential for confusion and ambiguity. With these issues in mind, I have designed a new thaaT system with a scientific underpinning, universalist approach, and algorithmic method. This system allows any raag to be unambiguously and uniquely classified into a thaaT based solely on the svar it allows, making this system far easier to understand and implement. In this article, I present a detailed description of this new thaaT system, contrasting it with the Bhatkhande scheme, and highlighting its relative strengths. A full *Table of ThaaTs* is given at the end of the article, with example raags that belong to each thaaT. Where an existing Hindustaanee raag couldn't be easily found, a Karnaatik raag or mela is given in italics.

2. Definitions and notation

2.1. Svar

We will define svar simply as a conceptually delimited pitch area, of which there are seven in an octave – Saa, Re, Ga, Ma, Pa, Dha and Ni. 2 of these – Saa and Pa – are much narrower pitch bands than the others, having no variants (no komal or teevra), and are often referred to as ‘achal’ svar. These are notated as S and P.

2.1.1. Shuddha: shuddha denotes ‘pure’ and shuddhataa means ‘purity’. Shuddha svar are considered more ‘natural’, and comprise the shuddha saptak, roughly equivalent to the Western major scale. 5 of the svar — Re, Ga, Ma, Dha and Ni — have 2 variants each, and are hence referred to as ‘chala’ svar. The variant having a more ‘natural’ or closer harmonic relationship with Saa is called the ‘shuddha’ variant of that svar. Shuddha Re, Ga, Ma, Dha and Ni are notated as uppercase R, G, M, D and N respectively.

2.1.2. Vikrit: The non-shuddha variant of a svar is called vikrit, meaning, ‘altered’; it could be komal or teevra.

2.1.3. Komal: komal is roughly equivalent to the Western term ‘minor’. The 4 svar Re, Ga, Dha and Ni have a komal variant each, which are very roughly 1 semitone below their respective shuddha variants. The komal svar are notated as lowercase r, g, d and n respectively.

2.1.4. Teevra: teevra is roughly equivalent to the Western term ‘augmented’. Only 1 svar – Ma – has a teevra variety, which is very roughly 1 semitone above shuddha M. Teevra M is notated as lowercase m.

2.2. Svarsthaan

Each variant of the seven svar is counted as 1 svarsthaan; thus there are 12 svarsthaan — 2 absolute (S, P), 5 shuddha (R, G, M, D, N), 4 komal (r, g, d, n) and 1 teevra (m). In common

parlance both terms are sometimes used interchangeably; however, strictly speaking, there are 7 svar, but 12 svarsthaan.

2.3. Raag

The foundational harmonic-melodic principle in Hindustaanee ‘classical’ or raagdaaree music, a raag can be conceptualized as a musical mood, a harmonic-melodic framework for composition and improvisation. At the most basic level, it comprises of a certain subset of svarsthaan (from the total 12) and a certain melodic grammar (i.e. the ‘chalan’) that dictates how and when these svarsthaans are to be used, how to move between them, etc. The ‘aaro-h-avaroh’ (simplest forms of ascending and descending scales that maintain the rules of the raag) is commonly used as an (oversimplified but indicative) outline to delimit and encapsulate a raag.

2.4. ThaaT / Mela

A *thaaT* can be defined as a parent scale having 2 characteristics — firstly that it has all the seven svar, and secondly that it has only 1 variety of each. Because a thaaT just an assemblage of svar, it is denoted by an ascending scale only — there is no descending scale. The term mela is the Karnaatik equivalent of the Hindustaanee thaaT; the system is called the melakartaa system.

3. **Need for any thaaT system at all**

Given the vast number of raags that exist, especially in online databases and books that include rare or ‘*aprachalit*’ raags, it is very useful to have some system or systems to classify them. As seen above, the two defining features of a raag are the svarsthaan it allows, and the chalan between them. It logically follows that there must be two primary classificatory systems, one for each of these features — the thaaT system, which will examine in detail, and the raagaang system, which is based on raag grammar and common phrases in raag ‘families’. Even if for no other reason, simply to organize raags in textual material, one still needs a system. I firmly believe that thaaT wins out as a more convenient technical classification, given its easily-understood and unambiguous rules, and its consistency and coherence. Wherever raags need to be sorted or ordered, whether in a database, book, syllabus, etc., the thaaT system proves its worth. An added benefit is that raags with the same svarsthaan are grouped together, which is very useful for instruments like the santoor that have a limited number of playable svar tuned to a particular scale — one can quickly look up all the raags that can be played in a given tuning without spending time retuning strings. For other instruments like the baansuree, vikrit svar are typically more difficult to play, especially for beginners. One can then choose raags that belong to thaaTs having none, and slowly graduate thaaTs with more vikrit svar.

4. **Need for a new thaaT system**

4.1. Problems with the Bhatkhande 10-thaaT system

The biggest failing of the Bhatkhande system is that it totally fails in classifying certain raags without recourse to the parallel raagaang system — Raag Aaheer-Bhairav is a great example of this. Since it has komal n and shuddha D, it doesn’t fit into the ‘Bhairav’ thaaT logically — but is forced into it based on raagaang similarity. Further limiting its effectiveness for raag classification is the Bhatkhande system’s ambiguity — some raags can be classified under multiple thaaTs. Raag Bhoopaalee is an oft-quoted example of this, which we will examine in detail in Section 7.3.3.

4.2. Problems with the melakartaa system

Though the melakartaa system used in Karnaatik raag music is far more comprehensive and rigorous, we cannot use it for Hindustaanee raag music, mainly because it uses different

svarsthaan concepts. This leads to some scales that have two variants of a svar and omit others altogether. Look no further than the very first mela, Kanakaangi, which in Hindustaanee notation has a scale of S r R M P d D S' — both R's and D's, and no G and N. Furthermore, the ethno-religious specificity of its chakra and mela names makes comprehension by musicians of non-Hindu and non-South Asian cultures, difficult and non-intuitive.

5. Overall structure of the new system

5.1. Total number of thaaTs

If we define a thaaT as a parent scale having all 7 svaras, with only 1 variety (whether shuddha or vikrit) of each, all possible combinations of shuddha and vikrit svar give us exactly 32 distinct scales. The total number of possible scales is calculated by multiplying the number of possibilities (shuddha or vikrit, thus 2) for each the svar (R, G, M, D, N) with variants. This gives us $2 \times 2 \times 2 \times 2 \times 2$ or $2^5 = 32$. Because S and P don't have variants, they don't generate any additional possibilities.

I have explained this calculation in detail to demonstrate how this number is beyond dispute or subjective preference, unless we change the definitions involved. As Lakshmi Sreeram¹ says, “Every mela will have 1 and only 1 variety of every note... So given that, and *given the way svaras are named and organized in Hindustani music, 32 combinations or 32 melas are possible.*” [Emphasis mine] This is important to me in that it proves the scientific basis for the foundation of my new thaaT system, and absolves it from accusations of having just another arbitrary number of thaaTs — historically, various numbers have been proposed, including 10, 11 etc. Sreeram goes on to acknowledge that “Bhatkhande knew the 72 mela of Karnaatik music. He also computed 32 *possible* melas for Hindustaanee music. ... But... he only listed 10 combinations... and he said these 10 are enough to account for or to classify all the known raags of that time.” According to her, he also acknowledged that others might feel the need for more thaaTs, and said that “that was fine.” Though I personally feel no need to draw on the authority of other musicologists, especially those far removed from the present, I trust these quotes will allay at least some hesitations of traditionalists to adopt a new system created by a nobody upstart like myself ☺.

5.2. Sequence

5.2.1. Having arrived at the number of thaaTs we will have, I needed to decide what sequence to place them in. Inspired by the Karnaatik *melakarta* system, I wanted to ensure that just by knowing the number or name of any *thaaT*, we can deduce its entire scale without any need to memorize each thaaT scale. I decided to start with the shuddha saptak, i.e. the scale with all svar in their shuddha form. This felt the most natural to me for a few reasons. First, the seven shuddha svar comprising this scale are the very reason we conceive of the ‘sapta sur’ in the first place — they are harmonically the closest to S and involve some of the simplest frequency ratios of all the typical musical notes. Second, in South Asia, this is the scale most beginners learn first, at least in the qhayaal tradition and modern school-based education system. Third, it is the scale most familiar to the typical non-musician, at least in many cultures of the world. And 4th, it is the

¹Sreeram, L., 2022. *Classification of Ragas - The Mela or Thaat System*; lecture as part of online course on ‘Appreciating Hindustani Music’ [Accessed 2023].

Available at: https://www.youtube.com/watch?v=Pj2xgyx1sLo&ab_channel=NPTEL-NOCIITM

most basic and ‘natural’ scale in the Western classical tradition as well, which is likely the most widely known music theory tradition known around the world, and also the one most dominating the contemporary world of music technology, software and non-traditional instrument manufacture.

5.2.2. To generate a full set of all 32 *thaaT* scales that include all possible permutations of *shuddha* and *vikrit svarsthaan*, I start with the *shuddha* variants of each *sva*, then, to get each subsequent scale, keep changing the variants of the 5 ‘*chala*’ *sva*.

5.2.3. The sequence in which the *komal svar* appear in each successive *thaaT* scale is N, G, D, and R. This results from the *komal svar* being ordered from ‘lightest’ to ‘heaviest’, in terms of their harmonic distance from *Saa*; i.e. from *Saa*, how many steps do we go ‘up’ by a ratio of 4/3 or *shadja-madhyam sambhaav* (perfect fourth interval) to get to that *sva*. [Thus they are sequenced in order of increasing harmonic distance from *Saa* in Pythagorean or 3-limit just intonation. The scientific reasoning for this sequence involves calculations of frequency ratios of *svarsthaan* and *sambhaav* (harmonic relation between 2 *sva*) — this is a complex and technical subject beyond this article’s scope, but more details are available upon request.] Thus, the N alternates between *shuddha* and *komal* in every *thaaT*; the G alternates between *shuddha* and *komal* every 2 *thaaTs*; the D every 4 *thaaTs*; the R every 8 *thaaTs*.

5.2.4. Since M is the only *sva* with a *teevra* variant, it is kept as the last changing *vikrit svar*. This has the advantage of neatly splitting the collection of *thaaTs* into 2 halves — all the *thaaTs* in the first half have a *shuddha* M, and all the *thaaTs* in the second half have a *teevra m*. This exactly mimics the *Karnaatik melakarta* system.

5.3. Grouping

5.3.1. Once I had this table of all the *thaaT* scales, again inspired by *melakarta* system (which has 12 *chakras* or groups of 6 *mela* each), I grouped together every 4 *thaaTs* into a *chakra*, creating a total of 8 *chakras*. Each *thaaT* within each *chakra* occupies a certain ‘position’ — the first, second, third or 4th.

5.3.2. Within each *chakra*, the ‘heavier’ R and D pair remain constant, i.e. maintain their particular permutation of *shuddha* and *komal*. Their cycle completes after 4 *chakras*, and repeats for the next 4 *chakras*. As the ‘lighter’ pair, the Ga and Ni permutations change from position to position within a *chakra*, and remain consistent at the same position in each *chakra*.

6. Naming

When it came to naming all the elements in this system, I decided to break with tradition drastically. The most important reason for this is that it is important to be inclusive and global in our music theory, to make it most friendly to all communities, and make it easily translatable into all the languages of the globe. Ideally, I wanted to use categories of easily-recognizable natural phenomena, whose numbers of member elements exactly matched the number of the elements in the *thaaT* system that I was naming after them.

6.1. Chakra names

For the names of the 8 *chakras*, colours suggested themselves to me readily. The only difficulty was that the widely-recognized 7 colours of the rainbow (VIBGYOR: violet, indigo, blue, green, yellow, orange and red) still left 1 *chakra* unnamed. For that, I added white, and put it at the top — white being a symbol for purity and peace, felt like a natural choice for the first *chakra*, which contains the first *thaaT* with only *shuddha svar*. White being the ‘hottest’ colour in terms of temperature also influenced this choice. The other colours follow in

decreasing order of temperature. This neatly mapped onto the chakras such that the chakras with shuddha M were named after the 4 warm colours, and the 4 cool colours became the names of the chakras with teevra m.

6.2. Position names

Since there are exactly 4 thaaTs in each chakra, the 4 cardinal directions of the compass were an easy choice for the positions – north, east, west and south, in that specific order, which the word ‘news’ derives from. This order is easy to remember using the word news, and it also aids in visualizing the G–N permutation. First, the 2 poles: north has G–N shuddha–shuddha, south has G–N komal–komal [I admit to a northern hemisphere bias in this, but again, the order in ‘news’ is my justification]. Second, east and west: for these, the cartographical convention of west being on the left and east being on the right indicates whether N or G is komal, according to left-to-right writing systems — because N is to the right of G in these writing systems, only N is komal in the east, and only G is komal in the west.

6.3. ThaaT names

For the thaaT names themselves, I decided to simply use a compound of their chakra and position, such as White-North (all shuddha svar), or Violet-South (all possible vikrit). This facilitates deducing the scale of the thaaT from its name. Now, many may wonder, “Why not use raag names for thaaTs, as Bhatkhande did?” I strongly believe that using raag names as thaaT names leads to confusion and should be avoided. First, thaaTs are merely conceptual constructs, created —for pedagogical, epistemological, logistical or technological purposes — to organize or classify raags using a systematic framework. Using raag names as thaaT names tends to subconsciously imbue thaaTs with raag-like properties, makes them more personified and personable, can blur the distinction between raag and thaaT, and make it easy for students to forget the vital distinctions between the two. On the other hand, *not* using raag names prevents build-up of any chalan or musico-emotional associations with thaaTs. The compound chakra-position names constantly remind us that thaaTs are musically sterile, mere scientific devices. They may also help remove the temptation to say, completely incorrectly and ahistorically, that Raag X ‘comes from’ or ‘is based on’ thaaT Y, which I hear so many musicians say — a pet peeve of mine. Second, raags change over time — its svars may change in relative importance, or may be dropped or added (e.g. Chandrakauns, Bihaag); even the raag itself may fall into disuse or disappear entirely. In such cases, we’re left with an anachronistic thaaT name that will not make sense, especially to beginners. [Just like many English language spellings don’t make sense because they stayed the same while the pronunciations changed.] Third, thaaTs are a fixed set of 7 svarsthaans, while raags can have additional svarsthaans, or can disallow some of the svarsthaans of the thaaTs they are classified into. Using raag names as thaaT names can lead to confusion about the svarsthaans that are allowed in a particular thaaT. Two examples: Raag Khamaaj includes shuddha N, whereas the Bhatkhande thaaT of that name has only komal n; Raag Maarvaa doesn’t have P, whereas the Bhatkhande thaaT of that name does. Third, it requires even experts to append ‘raag’ or ‘thaaT’ every time they use the name of a specific one — one can’t simply say ‘Bhairav’, without specifying each time whether one is talking of the raag or thaaT – though this may sound trivial, it creates an unnecessary burden, especially in lengthy verbal discussions. Overall, I believe that using distinct and separate names for thaaTs and raags can help to reduce confusion and provide greater clarity in the study and practice of Hindustani classical music.

7. Application

7.1. Deriving a thaaT scale from a thaaT name

Let us see how one can derive a thaaT scale from any given thaaT name. To take a random example, let's use Indigo-West. Right away we can see that indigo is a cool colour, so the M is teevra. Indigo being the 3rd of the cool-colour chakras (green, blue, indigo, violet), the R–D pair will have R komal and D shuddha (remembering that the 1st chakra in each half of the set will have R–D shuddha-shuddha, the second shuddha-komal, and the 4th komal-komal). Lastly, the position being west (on the left-hand-side on a map) indicates that only the G will be komal in the G–N pair. Thus we arrive at the scale S r g m P D N S', which corresponds to the Karnaatik mela number 47, Suvarnaangi, in the 8th chakra (Vasu).

7.2. Deriving a thaaT scale from a thaaT number

Let us see how one can derive a thaaT scale from any given thaaT number. To take a random example, let's use thaaT #13. Right away we can see that since the number is less than (or equal to) 16, the M is shuddha. To find the chakra number, we divide the thaaT number by 4, giving us a quotient of 3 and a remainder of 1. The quotient plus 1 gives us the chakra number – 4. The 4th chakra will have the R–D pair as komal-komal (remembering that the 1st chakra in each half of the set will have R–D shuddha-shuddha, the 2nd shuddha-komal, and the 3rd komal-shuddha). Lastly, the remainder of 1 indicates that this is the first thaaT in the chakra, i.e. in the north position, so both svar in the G–N pair will be shuddha. Thus we arrive at the scale S r G M P d N S', which matches the svar-set of Raag Bhairav, and corresponds to the Karnaatik mela number 15, MaayaamaalavagauLaa, in the 3rd chakra (Agni).

7.3. Rules for raag classification

7.3.1. Coming to the rules by which we classify raags under a particular thaaT in this new system. These rules are based solely on the svarsthaan used or allowed in the raag, and not on its raagaang, chalan, similarity to any other raag, or any other factor whatsoever. This provides a more objective classification system, removing subjective assessments of the relatively esoteric realm of raag chalan and grammar. What makes it all the more simple is that there are only 2 rules to consider. The first rule is that an absent svar will be considered shuddha. This means that if a raag does not use a certain svar (i.e. it is varjit), that svar will be assumed to be shuddha for the sake of classifying the raag under a thaaT, even if that (absent) svar is typically vikrit in other related or co-raagaang raags. The second rule is that if a raag uses both variants of a svar, it will be counted as vikrit, not as shuddha.

7.3.2. Let's take the example of Raag Vrindaavanee Saarang to demonstrate both rules in action. Because G and D are varjit, they are both considered shuddha. And since there are both N, the N will be considered vikrit. R and M are both shuddha anyway. So the thaaT of this raag becomes the scale S R G M P D n S' — i.e. White-East, thaaT #2, in chakra #1.

7.3.3. Let's take another example, that of Raag Bhoopalee, to see how this new system makes much more internal sense than the old Bhatkhande 10-thaaT system. Lakshmi Sreeram² shows how there could be 4 possible scale interpretations for Raag Bhoopalee (or Bhoop), given that it has 2 varjit svar. Out of these, 3 are thaaTs in the 10-thaaT system — Bilaaval, Khamaaj, Kalyaan — and the 4th doesn't exist in the Bhatkhande system but corresponds to the scale of Raag Vaachaspati. According to her, the performance tradition is clear – that Bhoop has to be classified under the Kalyaan thaaT due to the raagaang paddhati since the phrases and chalan are in common with Kalyaan or Yaman.

² Same lecture as in previous footnote.

In her own words, “We often have the cluster of notes, the same set of notes, but they give a very different personality. This can be accommodated under the raagaang paddhati, not under mela, obviously.” I argue that since it’s “obvious”... a) Since they’re clearly, completely different systems with different foci, why have raagaang-coloured expectations from thaaT, or vice versa? They’re totally separate, parallel systems, and should be used separately and purely, i.e. with no cross-contamination or confusion between the two. And b) Why use raagaang to justify arbitrary classifications in the thaaT system, like Bhoopaalee? If the Bhatkhande 10-thaaT system can’t classify a raag (or justify its classification under a particular thaaT) on its own merits and using its own internal logic, what’s the value of such a system? Whereas in my new system, simply applying the first rule automatically eliminates 3 of those 4 potential scales (the ones with komal n, with teevra m and with komal n *and* teevra m), leaving White-North (thaaT #1) as the only possibility.

8. Shuddhataa rank

8.1. Reasons for inclusion

For some, the ‘shuddhataa rank’ may be the most confusing part of my new thaaT system. So I’ll start with a strong disclaimer: the shuddhataa rank is NOT an essential part of the thaaT system, and can be safely ignored without affecting the classification of raags into the 32 thaaTs *in any* way. That being said, there are a few reasons I still included it in my table of thaaTs. First, since I used a computer program to generate the thaaT-scale database anyway, adding one more element with somewhat complex mathematical calculations involved was very easy. Second, and more importantly, I created this system with not just paper books and live students in mind, but especially also online databases, interactive websites, mobile phone apps, and the like. In this kind of database thinking, the more sorting criteria there are for tables of data, the better. Shuddhataa rank gives us an additional way to sequence the thaaTs that can be useful and helpful to many, as I shall show, and yet creates no additional complications for anyone who doesn’t want to use it.

8.2. Benefits

First and foremost, ordering the thaaTs by increasing number of vikrit svar helpful in a number of ways. On many instruments, like the baansuree, the guitar and piano in certain keys, and even for many beginner singers, vikrit svar (especially komal) take more effort and technique to produce and sustain. Hence having ordered by shuddhataa rank gives them a sequence of thaaTs in increasing order of difficulty or complexity, which they can work on successively as their musical skills improve. Second, while ordering thaaTs by thaaT number is great for being able to derive their scales from their numbers, it doesn’t always match our musical instinct; the shuddhataa rank can do better on this score. For example, it places Green-North (thaaT #17, or the old ‘Kalyaan’ thaaT in Bhatkhande) right after White-North (thaaT #1, or the old ‘Bilaaval’ thaaT in Bhatkhande), which may make more intuitive and harmonic sense to some.

8.3. Methodology and calculation

To arrive at the actual figures for each thaaT’s shuddhataa rank, I started with defining that White-North (thaaT #1) would have a rank of 0, following the convention of lower numbers being higher ranks, and given that the scale with only shuddha svar has the highest level of shuddhataa a thaaT scale can have. On the opposite end of the spectrum, Violet-South (thaaT #32) had to have a rank of 100 by definition, since it had the maximum possible number of vikrit svar. Next, I gave each vikrit svar a specific ‘weight value’, following the same ‘lightest’ to ‘heaviest’ order for the komal svar discussed in section 5.2.3, with the important distinction that teevra M was given the least weight of all vikrit svar, again for reasons of both Hindustaanee musical instinct and the

physics of the harmonic overtone series. Lastly, having X number of vikrit svar (0 to 5) itself was made to increase the shuddhataa rank. Then the program just added together the 2 quantities — number of vikrit svar and weighted values of each specific vikrit svar contained — to arrive at the final shuddhataa rank for each thaaT seen in the Table of ThaaTs.

8.4. Applicability to raags

It may seem that the numbers given as shuddhataa ranks are unnecessarily complex, going up to 6 decimal places and all. This was deliberate — it enables the shuddhataa ranking system to be applied to raags as well, not just thaaTs. The high precision of these numbers, resulting from the highly specific, NON-round-figure values taken for vikrit svar weighting, will hopefully ensure that each raag will have a unique shuddhataa rank. This will be especially useful in online databases of all existent raags. [The exact calculations and even the program patch used to generate the whole system are available upon request.]

9. **Potential criticisms and their refutations**

9.1. Ignoring other musical factors

The most typical criticism I anticipate this system getting, is that it totally ignores raag grammar, chalan, raagaang and the like. My rejoinder to this is that it is an invalid critique of my new thaaT system, simply because it blames the system for not accomplishing something it was never built to do in the first place! As I've said right in the Introduction, the system only looks at svarsthaan used in a raag, and is completely agnostic to other factors. It is merely ONE of the many ways in which we can classify raags, and in an ideal database, especially digital, all these other classification systems will be used in tandem, and a user can sort by or search for raags using any or all of these criteria.

9.2. Reduces raags to scales

The second-most common refrain is that since thaaT ignores all these vital musical aspects of raag, it should not be used at all, as it leads to a watering-down of raag subtleties and spurs the oversimplification of raags, thus harming our Hindustaanee classical tradition. Without reiterating the arguments I made immediately above, my very simple response is that any musician who feels that thaaT is a useless lens to view raags through, is totally free to ignore all thaaT systems completely, and not engage with or use or refer to thaaT at all, whether in performance or pedagogy! This new system was designed especially as a *replacement* for the problematic Bhatkhande 10-thaaT system — so it should be used wherever that system is already being used, and in new spaces like online databases as mentioned above. It does NOT seek to replace or even compete with other classification systems of raag, — raagaang, jaati (penta/hexa/hepta-tonic), etc.

9.3. Don't need so many thaaTs

We have seen how some justify Bhatkhande choosing only 10 out of the possible 32 thaaTs by pointing to the lack of raags which have the svarsthaan set of some of the other “possible” thaaTs. Their argument is that if there is no raag classifiable under a particular thaaT, there's no need to have that thaaT at all. I firmly refute this. First, a system of classification should be complete and whole, following its internal logic to the end, not a stopgap measure to cover current circumstances only. I would especially point to the analogy of the Periodic Table of Elements, arranged in columns and rows according to atomic number and chemical properties, and leaving spaces for undiscovered elements, many of which were later discovered, sometimes aided by the predictions of the periodic table itself! Hence to leave out svarsthaan permutations just because they aren't in contemporary raag usage seems unwise to me. Second, new raags are being created all the time [including by me ☺]. And further, let's not forget that all the thaaTs in this new

system that MAY not have any Hindustaanee raags to claim as their own, *right now*, still exist as Karnaatik melakartaa raags already! It may only be a matter of time before they are imported into the Hindustaanee pantheon like so many before them - Hamsadhvani, Keervaanee, Simhendramadhyamam, Vaachaspati... The list goes on and on. Third, referring to the Table of ThaaTs shows that I could not find Hindustaanee raags for *only 5 out of 32* thaaTs (indicated by the melas in italics for them) — and this was after only cursory Internet searches... true aprachalit raag experts may indeed be able to name an example raag for each thaaT! [This excludes Karnaatik raags already performed by established Hindustaanee musicians.] So clearly this argument is worthless; at the minimum, it can't be used to defend having *only 10* thaaTs!

9.4. Inappropriate naming

Some critics may argue that using colour names and cardinal directions to describe musical scales is not appropriate or relevant to music theory, and that some languages may not have distinct names for all the colours or directions used in the system, which could potentially cause confusion or difficulty in learning and using the system. I accept that these terms are not *completely* universal, but would counter that this is definitely a far more culturally inclusive approach to naming than the melakartaa system's chakra names, which are derived from Hindu mythology. I contend that these 7 colours and 4 directions exist, *at least* as concepts, in all the most-spoken languages of populations with significant proportions engaged in Hindustaanee raag music — certainly in South Asia, Europe and the Americas. And my final argument is that the names of the chakras, the positions, and thus of the thaaTs themselves, are all completely replaceable by other names, if that makes this system more acceptable to practitioners and teachers of Hindustaanee raag music. The crux of the new system is the number of thaaTs, with the sequence coming second in importance, and the names not at all affecting the integrity and working of the system.

9.5. Too complex

Finally, some may argue that the new system may be too complex for some learners, especially beginners, to understand and memorize. I would counter that by saying that a) beginners should not be asked to memorize the entire table of thaaTs! The need for classification of raags arises only once one is dealing with a certain number of raags, and by then, one isn't a beginner! And b) though this *article* gives a lot of information about the calculations of different svarsthaan permutations, and references mathematical algorithms for calculating the shuddhataa ranks, all that is only for someone trying to understand the theoretical underpinnings of the new system. Anyone *using* the thaaT system doesn't need to bother about any of that. They can just look at the Table of ThaaTs, which contains all the information needed to apply the system or assign a raag to a thaaT.

10. Conclusion

When asked about my innovation in raag classification, well-known qhayaal vocalist Shatavisha Mukherjee had this to say: "This new thaaT system and explanation opens a whole new door to the appreciation and practice of Hindustaanee classical music. It compels musicians to think in a new way, which is helpful for the betterment of our own understanding of music. His usage of universal terms and mathematical equations make this system a globally relatable one. I hope soon this system will change the perspective of the thaaT tradition."

My new system can be used to classify not only raags, but also Western scales, maqaams, dastgahs, and other musical modes. Any music system that has 7 notes (and variations thereof, even if more than 1) will be amenable to this. With this system, Hindustaanee musicians can communicate with musicians from other cultures more effectively by using a common language to

bridge the cultural divide. Thus the system facilitates cross-cultural communication and collaboration. And for Hindustaanee raagdaaree composer-performers themselves, the new system encourages and facilitates the discovery / creation of new raags, besides helping understand relationships between existing ones in a new and consistent way. Overall, I feel that my new thaaT system offers a more comprehensive and reliable framework for understanding and organizing the rich diversity of Hindustaanee raags in all their glory.

Table of ThaaTs

CHAKR A NUMBE R	CHAK RA	POSITIO N	THAAT NUMBER	SCALE	SHUDDHATA A RANK	EXAMPLE RAAG
I	White	North	1	S R G M P D N S'	0	Deshkaar
		East	2	S R G M P D n S'	16.793893	Kaamboji
		West	3	S R g M P D N S'	18.320611	Aabhogi
		South	4	S R g M P D n S'	35.114504	Jaijaivanti
II	Red	North	5	S R G M P d N S'	21.374046	NaT Bhairav
		East	6	S R G M P d n S'	38.167939	Chaarukeshi
		West	7	S R g M P d N S'	39.694657	Chandrakauns
		South	8	S R g M P d n S'	56.48855	Peeloo
III	Orange	North	9	S r G M P D N S'	27.480916	BhaTiyaaree Bhairav
		East	10	S r G M P D n S'	44.274809	Bairaagi
		West	11	S r g M P D N S'	45.801527	Kokil Puriyaa (?)
		South	12	S r g M P D n S'	62.59542	Salagvaraalee ToDi
IV	Yellow	North	13	S r G M P d N S'	48.854962	Gunkali
		East	14	S r G M P d n S'	65.648855	Basant Mukhaari
		West	15	S r g M P d N S'	67.175573	Bhoopaal ToDi
		South	16	S r g M P d n S'	83.969466	Bilaaskhaanee ToDi
V	Green	North	17	S R G m P D N S'	16.030534	Bihaag
		East	18	S R G m P D n S'	32.824428	Sarasvati
		West	19	S R g m P D N S'	34.351145	Madhuvanti
		South	20	S R g m P D n S'	51.145038	Madhukauns
VI	Blue	North	21	S R G m P d N S'	37.40458	Lataangi
		East	22	S R G m P d n S'	54.198473	<i>Rishabhapriyaa</i>
		West	23	S R g m P d N S'	55.725191	Simhendramadhyamam
		South	24	S R g m P d n S'	72.519084	<i>Shanmukhapriyaa</i>
VII	Indigo	North	25	S r G m P D N S'	43.51145	Sohni
		East	26	S r G m P D n S'	60.305344	Priya Kalyaan
		West	27	S r g m P D N S'	61.832061	<i>Suvarnaangi</i>
		South	28	S r g m P D n S'	78.625954	<i>ShaDvidamaargini</i>
VIII	Violet	North	29	S r G m P d N S'	64.885496	Shree
		East	30	S r G m P d n S'	81.679389	<i>NaamanaaraayaNi</i>
		West	31	S r g m P d N S'	83.206107	Gurjari ToDi
		South	32	S r g m P d n S'	100.000000	Adaarangee ToDi